

Practise Question

Solve all the question (paper or any software) and upload it in MS Teams on or before 20th

Sept 2020, 5:00pm

1. Apply SJF scheduling algorithm with 6 processes the processes and their respective Arrival and Burst time are given in the following table. Draw the Gantt chart and find the turnaround time (TAT), waiting time (WT), Avg. waiting time (AWT), Process completion time

Process	Arrival Time	Burst Time
1	1	7
2	3	3
3	6	2
4	7	10
5	8	8
6	9	2

2. Apply Shortest Remaining Time (SRTF) scheduling algorithm with 6 processes the processes and their respective Arrival and Burst time are given in the following table. Draw the Gantt chart and find the turnaround time (TAT), waiting time (WT), Avg. waiting time (AWT), Process completion time

Process	Arrival Time	Burst Time
1	0	8
2	1	4
3	2	2
4	3	1
5	4	2
6	5	3

3. Apply Shortest Remaining Time (SRTF) scheduling algorithm with 3 processes the processes and their respective Arrival and Burst time are given in the following table. Draw the Gantt chart and find the turnaround time (TAT), waiting time (WT), Avg. waiting time (AWT), Process completion time

Process	Arrival Time	Burst Time
1	0	9
2	1	4

3	2	9
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4. Apply non pre-emptive SJF scheduling algorithm with 4 processes the processes and their respective Arrival and Burst time are given in the following table. Draw the Gantt chart and find the turnaround time (TAT), waiting time (WT), Avg. waiting time (AWT), Process completion time

Process	Arrival Time	Burst Time
1	0	21
2	1	3
3	2	6
4	2	2

5. Apply pre-emptive SJF scheduling algorithm with 4 processes the processes and their respective Arrival and Burst time are given in the following table. Draw the Gantt chart and find the turnaround time (TAT), waiting time (WT), Avg. waiting time (AWT), Process completion time

Process	Arrival Time	Burst Time
1	0	12
2	1	31
3	2	16
4	2	18

6. Apply non pre-emptive SJF scheduling algorithm with 5 processes the processes and their respective Arrival and Burst time are given in the following table. Draw the Gantt chart and find the turnaround time (TAT), waiting time (WT), Avg. waiting time (AWT), Process completion time

Process	Arrival Time	Burst Time
1	2	6
2	5	2
3	1	8
4	0	3
5	4	4

7. Apply pre-emptive SJF scheduling algorithm with 5 processes the processes and their respective Arrival and Burst time are given in the following table. Draw the Gantt chart and find the turnaround time (TAT), waiting time (WT), Avg. waiting time (AWT), Process completion time

Process	Arrival Time	Burst Time
1	0	8
2	5	12
3	1	8
4	0	13
5	4	4

8. Apply non pre-emptive SJF scheduling algorithm with 5 processes the processes and their respective Arrival and Burst time are given in the following table. Draw the Gantt chart and find the turnaround time (TAT), waiting time (WT), Avg. waiting time (AWT), Process completion time

Process	Arrival Time	Burst Time
1	2	3
2	0	4
3	4	2
4	5	1
5	4	4

9. Apply pre-emptive SJF scheduling algorithm with 5 processes the processes and their respective Arrival and Burst time are given in the following table. Draw the Gantt chart and find the turnaround time (TAT), waiting time (WT), Avg. waiting time (AWT), Process completion time

Process	Arrival Time	Burst Time
1	3	1
2	1	4
3	4	2
4	0	6
5	2	3

10. Apply non pre-emptive SJF scheduling algorithm with 5 processes the processes and their respective Arrival and Burst time are given in the following table. Draw the Gantt chart and find the turnaround time (TAT), waiting time (WT), Avg. waiting time (AWT), Process completion time

Process	Arrival Time	Burst Time
1	2	11
2	1	8
3	5	2
4	0	16
5	3	3

NOTE :

1. Completion Time (CT) = Time at which process completes its execution.
2. Turnaround Time (TAT) = Completion Time (CT) - Arrival Time (AT)
3. Waiting Time (WT) = Turnaround time (TAT) - Burst Time (BT)
4. You can use the table mentioned below to solve the questions.

Gantt Chart

P0	P1	P2	..
0	3	8	14 ...

Process ID	Burst Time (BT)	Arrival Time (AT)	Completion Time (CT)	Turnaround Time TAT = CT - AT	Waiting Time WT = TAT - BT
X	X	X	X	X	X
X	X	X	X	X	X
..	..	..	..	..	..
Average Waiting Time =					
Average Turnaround Time =					